



Published in final edited form as:

Contemp Clin Trials. 2025 May ; 152: 107866. doi:10.1016/j.cct.2025.107866.

Monitoring in Pragmatic Trials Lessons from the NIH Pragmatic Trials Collaboratory

Lesley H. Curtis^a, Stephanie Morain^b, P. Pearl O'Rourke^c, Karen Staman^d, Jeffrey G. Jarvik^e, Andrea Cheville^f, Dana L Dailey^g, Kathleen A. Sluka^h, Patrick Heagertyⁱ, Edward R. Melnick^j, Hrishikesh Chakraborty^k, James A. Tulsky^{l,m,n}, Angelo Volandes^{l,o,p}, Gregory E. Simon^q

^aDuke University School of Medicine, Department of Population Health Sciences Durham, NC

^bJohns Hopkins Bloomberg School of Public Health, Johns Hopkins Berman Institute of Bioethics, Baltimore, MD USA

^cHarvard Medical School (Retired), Boston, MA, USA

^dDuke Clinical Research Institute, Durham, NC

^eDepartment of Radiology and Clinical Learning, Evidence, and Research (CLEAR) Center, University of Washington, School of Medicine, Seattle WA

^fMayo Clinic Comprehensive Cancer Center, Rochester, MN

^gDepartment of Physical Therapy, St. Ambrose University, Davenport, IA and Department of Physical Therapy and Rehabilitation Science, Carver College of Medicine, Iowa City, IA

^hDepartment of Physical Therapy and Rehabilitation Science, Carver College of Medicine, Iowa City, IA

ⁱUniversity of Washington, School of Public Health, Seattle WA

^jYale School of Medicine, Department of Emergency Medicine, New Haven CT

^kDuke University School of Medicine, Department of Biostatistics, Durham, NC

Address for correspondence: Lesley Curtis, Professor in Population Health Sciences, Duke University School of Medicine, 215 Morris St, Suite 210, Durham, NC 27701, Duke Box 104023, Durham, NC 27710, lesley.curtis@duke.edu.

Publisher's Disclaimer: This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Author contributions: CRediT

- Conceptualization: LC, SM, PO, GS
- Funding acquisition: LC
- Writing – original draft: LC, SM, PO, GS, KLS
- Writing – review and editing: LC, SM, PO, GS, KLS, JGJ, AC, DLD, KS, PH, ERM, HC, AV, JT

Disclosures

All authors declare that they have no financial interests related to the content of this work.

Declaration of Interest Statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

^lHarvard Medical School, Boston MA

^mDepartment of Supportive Oncology, Dana-Farber Cancer Institute, Boston, MA, Dana-Farber Cancer Institute, Boston, MA

ⁿDepartment of Medicine, Brigham and Women's Hospital, Boston, MA

^oDepartment of Medicine, Massachusetts General Hospital, Boston MA

^pACP Decisions, Boston, MA

^qKaiser Permanente Washington Health Research Institute, Seattle, Washington

Abstract

The distinguishing characteristics of pragmatic clinical trials merits special attention when developing a monitoring plan. Pragmatic clinical trials are large in scope; participants are often identified from records or routinely collected data; investigators typically have less control over treatments or interventions; outcome data are often extracted from health records; and study activities are commingled with usual health care. We use lessons from The NIH Pragmatic Trials Collaboratory, which supports the conduct of 32 pragmatic clinical trials, to illustrate some of the challenges and solutions. Challenges include the complexity, quality, and timing of a real-world data pipeline; interventions that are embedded in clinical workflows; and the potential for incidental findings. We recommend regular, rigorous data quality checks, ongoing monitoring of adherence to interventions, and including someone who is knowledgeable about pragmatic clinical trials and novel research designs in the development of Data and Safety Monitoring Plans and Data and Safety Monitoring Boards. Close monitoring by study leaders, independent monitors or and Data and Safety Monitoring Boards is critical for a successful study that produces meaningful results. These experts must also decide about what evidence requires action and/or modification of the protocol and what information and thresholds would lead to a decision to pivot or terminate the trial.

Keywords

monitoring; ethics; pragmatic clinical trials; safety

Introduction

Monitoring plans for clinical trials are required to support the integrity and safety of research conduct, including the collection and generation of research data. NIH policy requires an independent Data and Safety Monitoring Board (DSMB) for trials with a heightened risk to *individual* participants [1]. However, many trials can be safely monitored under a Data and Safety Monitoring *Plan* (DSMP) executed by either a single external monitor or scheduled review by designated study staff (Table 1). Even PCTs not involving heightened risk to individual participants may benefit from independent external monitoring of complex interventions or complex sources of study data. Pragmatic clinical trials (PCTs) may require some nuanced considerations for monitoring because they are embedded in one or often several dynamic and continually changing healthcare systems[2-5]. The laudable goal of

PCTs to reflect real-world conditions comes with the challenges of conducting research in settings that cannot be closely controlled and where research activities are commingled with ordinary health care.

Simon et al [6] previously articulated a list of what appropriate monitoring should include, and we have revised the list to be more specific to PCTs. While much of this list is applicable to all research, there are considerations for each item that are specific to pragmatic clinical trials, as we will discuss below.

1. Monitoring recruitment to assure enrollment of a sample sufficient to answer the study question
2. Monitoring safe and effective conduct of interventions delivered to study participants
3. Assuring that there is a plan to address potential incidental findings that are actionable, especially indicators of urgent clinical need [7]
4. Monitoring the timeliness and quality of study data
5. Monitoring outcome data for evidence that significant benefits or risks have been clearly demonstrated, or evidence that a meaningful difference in outcomes will not be observed (futility) [6].

This list is used to support the goal of safe and efficient research that produces reliable scientific evidence while respecting participant and clinician time as well as the resources required for that research [8].

Relevant Characteristics of PCTs

The NIH Pragmatic Trials Collaboratory [9,10] supports the design and conduct of 32 pragmatic trials embedded in healthcare systems across 49 states, involving >1400 clinical sites, >1.2 million patients, and 14 NIH institutes and centers. In our experience, monitoring of these PCTs has surfaced challenges that mirror the list above yet have their own particularities. For example, monitoring may identify unexpected challenges dealing with the heterogeneity of sites across academic centers and research-naïve sites; variability in site-buy-in; changing standards of care; and different and changing information systems.

PCTs differ from traditional trials in several ways. One critical difference is that they are conducted as part of routine care, and interventions are frequently delivered by frontline staff instead of research staff. Data may be collected from a variety of existing sources including, for example, the electronic health record, insurance claims, death records, as well as patient-reported outcomes. Thus, data sources are heterogeneous, and it can be difficult to validate independent, external systems. Timeliness of reporting can also be an issue, especially when data sources are pulled in batches and at specific time periods (monthly, every 3 months, etc.) from information systems that are continually being updated to serve clinical care. These features can complicate assessment of adverse events, affecting the ability to evaluate whether events are related to the intervention, given both that access to data tends to be limited or delayed, and that study staff may have limited or no direct contact with participants.

Additionally, the scale of PCTs tends to be large, involving multiple heterogeneous sites and thousands of patients across a variety of settings. The heterogeneity of sites poses unique challenges, as there may be important differences in the demographics and health of the populations served as well as in the sites themselves. A study may include research naïve sites, or sites that may have less research experience than highly resourced sites, making them less familiar with monitoring and reporting events. For these sites, monitoring may require additional support.

In this paper, we describe case studies from the NIH Collaboratory (Table 1) to illustrate challenges uncovered through monitoring and the solutions used to overcome the challenges. Key challenges and recommendations are shown in Table 2.

1. Monitoring recruitment to assure enrollment of a sample sufficient to answer the study question

All clinical studies must enroll a sufficient number of participants to yield results that are credible and reliable [6]. In PCTs, patients may be enrolled directly from the electronic health record. In this situation, it is important to monitor that the electronic health record system at each site provides adequate representation of participants, and that the enrolled participants are representative of people impacted by the condition under study. This monitoring is especially important in light of known biases in both EHR data capture and differential access to care for population subgroups [11]. Here, monitoring can be used to detect whether characteristics of participants, such as race and ethnicity, are accurately and completely captured at each site to help ensure a generalizable population is included. Early signals that enrollment is insufficient for subgroups can be countered with the addition of study sites, changes in recruitment strategies or materials, increased engagement with partners, or other changes to the protocol.

If a DSMB is involved, Zahrieh et al recommend considering how one would summarize accrual and baseline characteristics across the hierarchy of sites, clinicians, and patients, and if imbalances occur, create a plan for discerning barriers to accrual or for determining if the intervention is systematically disadvantaging certain demographic groups [5]. For example, during the planning phase, EMBED [12,13] researchers found that coding practice related to patient race and ethnicity varied not only from system to system but also within systems (See Table 1 for a description of all trials). This raised two important threats for PCTs: 1) Inaccurate coding of race/ethnicity, which could undermine validity, particularly if analyses involve comparisons by race/ethnicity; and 2) inaccurate or incomplete capture of race/ethnicity data, which could undermine the ability to assess whether data will, in fact, be representative of the populations for whom the research aims to inform future care practices and decision-making. Repeated, careful, and rigorous checks were needed to ensure fidelity of the data collection process during the planning phase. To monitor for this, EMBED researchers increased the number of data pulls during the trial.

The embedded nature of pragmatic trials naturally leads to clustering of patients within providers and facilities. Appropriate statistical methods are necessary to account for such clustering. Trial planning requires estimates of intra-cluster correlation coefficients (ICCs) which are frequently minimally informed from pilot or prior data. When necessary, trials

can conduct interim sample size reassessment focusing on estimates of variance or ICCs to ensure that the planned sample size is sufficient for inference. For example, FM-TIPS [14] conducted a formal interim evaluation of the within-clinic ICC and ultimately adjusted the participant recruitment goal from an initial target of 600 to a final target of 450 subjects. When imbalances are severe enough to affect statistical power, the DSMB can help with determinations about early termination of the trial or sample size re-estimation [5].

2. Monitoring safe and effective conduct of treatments delivered to study participants

Monitoring for *safe conduct* involves the detection of adverse events, as interventions that are presumed to be helpful may have unintended consequences in the real world. PCT investigators often have little or no direct contact with participants, and PCTs do not typically collect detailed clinical data to detect adverse events, so investigators should *a priori*, consider how unintended consequences could be detected. For example, LIRE was a stepped-wedge, cluster-randomized pragmatic clinical trial that tested the insertion of benchmark prevalence data into routine radiology reports for people who had received lumbar spine imaging [15,16]. The trial used data from the electronic health record and other information systems to identify clinics and primary care clinicians, insert intervention text into the radiology reports, tailor intervention text by imaging modality and patient age, verify intervention implementation, and assess outcomes.

Recognizing that the intervention could reduce detection of rare but potentially dangerous causes of back symptoms, external safety officers monitored for emergency room visits within 90 days and deaths within six months of index imaging [17].

Monitoring for *effective conduct* of treatments delivered to participants also requires observation of the fidelity or quality of treatment delivery [6] by monitoring adherence to the protocol or other quality parameters. This can be particularly difficult when the study includes multiple heterogeneous sites. A lack of monitoring could impede appropriate interpretation of the study results, as a failure to find a difference in treatment arms could be due to lack of efficacy *or* failure to adhere to interventions [4]. The Pain Management Collaboratory recommends early and regular adherence/fidelity monitoring to determine whether intervention delivery is as intended, as well as the development of thresholds for intervening [18]. If a DSMB is involved, the DSMB could focus on interim monitoring for futility or effectiveness of the intervention, as long as the monitoring will not affect the results in a substantive way [5]. Regardless of whether a DSMB is involved or not, when poor adherence is noted, study teams can implement ways to increase adherence, such as plan-do-study-act cycles [19], re-training, or by engaging with partners, including patients, health systems leaders, and front-line clinicians. Because PCTs aim to assess real-world effectiveness, efforts to increase adherence or fidelity should not exceed those that would be practicable in the settings where trial results would be applied.

Using the LIRE example, the EHR worked well for delivering the clinical decision support intervention, which was automated and customized to imaging modality and patient age. There was, however, implementation heterogeneity among sites. One site preferred to insert interventions as pop-up alerts that clinicians received, which were not permanently attached to the report. Some inserted intervention text using the radiology reporting system,

which allowed the dictating radiologist to remove intervention text if desired. To monitor implementation of interventions, the research team used regular data pulls to ensure text was inserted appropriately into the radiology reports. This led to the discovery that an upgrade to the dictation software at one of the sites had broken the link that inserted the intervention text, causing unintentional crossover. As a result, the frequency of monitoring was increased.

As another example, NOHARM is an ongoing study that aims to provide validated nonpharmacologic pain care options, while attenuating opioid use, in patients undergoing qualifying surgeries [20]. The electronic health record facilitates delivery of 3 intervention components. First, a conversation guide is automatically sent to the patient portal that describes opioid-related risks, the best practice of combining analgesic approaches, and the efficacy of nonpharmacologic modalities. The guide educates patients on 13 validated nonpharmacologic pain care options and allows them to enter selections that are saved as discrete EHR data elements. Second, a collection of clinical decision support tools prompts clinicians to support patients' use of their preferred modalities. Last, a suite of self-management tools enables patients to use and adapt the modalities for maximal benefit. The trial used the electronic health record to identify participants, support transitions as clusters went live with the intervention, deliver tailored messaging, prompt care teams, and collect outcomes.

Patient interaction with the portal guide was monitored to ascertain if they made and changed nonpharmacologic pain care selections. Inpatient nurses' delivery of education on patients' selected modalities and provision of the modalities to manage pain was monitored with regard to fidelity and safety. Nurses' safe clinical use of nonpharmacological modalities was tracked in the EHR with ICD code automated surveillance.

3. Assuring that there is a plan to address potential incidental findings, clinical needs, or other actionable signals of distress

PCT investigators or other study staff may discover a range of information types that may relate to the clinical needs of enrolled patients—or even others within the broader health system. Since pragmatic research is embedded in routine care, the boundaries of research and clinical care can be blurred [21-23], raising questions about the normative obligations of investigators and others within the health system to respond to signals of potential risk [7]. According to one influential account of research ethics, the obligations of researchers to participants are more limited than those of physicians to patients, and thus, the obligations of researchers to respond to signals of potential clinical need are more limited in both scope and scale than they would be in the context of clinical care. Such accounts, however, are complicated in the PCT context, in which research is deliberately integrated into clinical care, and in which individuals enrolled may expect to receive care for any health-related issues whether or not discovered as part of the research. Additionally, for many PCTs, a waiver of consent is granted, and therefore participants may not know they are in a research study, complicating decisions related to the ethical management of incidental and collateral findings. The large scale of PCTs may also require considerable effort and costs for disclosure and follow-up [21]. Finally, some findings may have implications for the well-being of non-subjects, including signals of broader care gaps within the participating

health systems [24]. In the planning phase, the IRB should help determine the events that need to be considered, the mechanisms for communication, and the responsible parties [7]. For monitoring, determining appropriate responsibilities will vary based upon features of a given PCT and the participating health systems. If a DSMB is involved, DSMB members could potentially help investigators review the adequacy of these plans over time, especially in light of unexpected events.

DSMB members experienced in traditional clinical trials often focus on whether the intervention itself is the cause of a clinical need or adverse event: this can be difficult to determine in PCTs due to the large number and heterogeneity of sites and individuals enrolled, delays in receiving data from different sources (i.e., claims data), and the fact that the research staff often have no interaction with individuals [21]. When designing a trial, there are many factors to consider for responding to a signal of potential patient risk, including the expectations of partners (patients, clinicians, and health systems leaders); the specific characteristics of the study population; how to define triggers, thresholds, and responsibilities for action; identifying appropriate response mechanisms and capabilities; integrating with clinical practices and systems, and protecting the participants' privacy [7]. While the likelihood of discovering an urgent clinical need may vary by study arm (based, on, for example, differential contact for those randomized to intervention versus control conditions), the threshold for action and subsequent response would typically be the same for all participants.

For example, FM-TIPS (Fibromyalgia TENS in Physical Therapy Study) is being conducted to examine if adding transcutaneous electrical nerve stimulation (TENS) to routine physical therapy treatment improves movement-evoked pain for patients with fibromyalgia [25]. FM-TIPS is part of the Helping End Addiction Long-Term (HEAL) initiative [26], which requires the collection of certain patient-reported outcome (PRO) measures for research purposes, including questionnaires on depression, anxiety, and substance use. The TENS unit is provided by the physical therapists (PTs) in the clinic and sent home with the participant. The PROs are collected remotely through REDCap®. Although a member of the study team monitors the data for signals of potential distress, data are only pulled monthly. As in many PCTs, the intervention itself is low risk (provision of a TENS unit). However, the characteristics of the population, in which depression and anxiety are common, merited special considerations. In this case, the clinic PTs serve on the front-line to help patient-subjects obtain necessary evaluation and care for potential mental health problems, with the study team providing periodic review through the monthly data checks [7].

4. Monitoring the timeliness and quality of study data

Data in pragmatic clinical trials are typically extracted from records generated by routine health care operations, and as such, researchers do not control the chain of custody of the data [6], and the data pipeline is more complex than in traditional clinical trials [2]. Thus, there are many potential sources of error that can affect data quality, including changes in the electronic health record, changes in workflow or business processes, and changes in the standard of care. Because integrity of the data can threaten the integrity of the overall study, Bunning et al argue that a DSMB should be required even in minimal risk studies to create

reports that describe the validity of the data driven aspects of enrollment, intervention uptake and delivery, and data collection [2].

ACP PEACE is an ongoing pragmatic stepped-wedge cluster-randomized controlled trial testing a combined intervention of video decision aids and communication skills training to improve advance care planning [27]. Investigators planned to use structured data elements from the EHR to examine advance care planning documentation. However, a chart review of data elements during the planning phase revealed that these were inaccurate—fewer than 60% had the correct documentation of advance care planning. ACP PEACE had the advantage of having a laboratory that leverages natural language processing, which made it possible to extract the same data with far more accuracy and speed. The study team recommends checking and validating data early, hiring and funding experts in IT, and having champions at each site to establish relationships with people who can help rectify any issues.

5. Monitoring outcome data for evidence that significant benefits or risks have been clearly demonstrated, or evidence that a meaningful difference in outcomes will not be observed (futility)

For a PCT or a traditional trial, if significant risks to patients are identified, the trial should be discontinued. The study may also be stopped if an intervention is found to be beneficial at the required level of statistical significance. For example, a pragmatic trial comparing video versus direct laryngoscopy trial was stopped for efficacy at the interim analysis [28]. However, for pragmatic clinical trials, even when significant benefits are identified early, careful consideration should be given to the amount of evidence required to drive actionable change in care. Health systems leaders make decisions using the best available information, including health system priorities, costs, opinion of colleagues and staff, and other factors, and therefore may not require nor respond to evidence that meets the nominal $P < 0.05$ level of significance required for the study [29]. As such, engagement with partners (patients, clinicians, and health systems leaders) before the launch of the study could be helpful as part of DSMB deliberations for determining thresholds and triggers that will guide early termination decisions.

If problems are identified that impact the integrity of a trial such that the trial is unlikely to achieve sufficient precision or sample size, trial investigators might first consider changes to the protocol that allow for a pivot such that the significant investment in the trial—which includes patient time, money from funders, researcher and clinician time, and opportunity costs—can still lead to evidence that is useful. For example, the HiLo trial changed the design from cluster-randomized to individually randomized due to differences in recruitment based on the intervention assignment of “Hi” serum phosphate target (≥ 6.5 mg/dL) or the “Lo” serum phosphate target. When a pivot is not successful, then termination of the trial is an option of last resort.

Conclusion

Even when a PCT is deemed minimal risk and does not require oversight from a DSMB, independent oversight from a safety officer or DSMB may be helpful to monitor the multi-

level data—collected from patients, providers, and practices—that is collected in real-world settings.

Lessons from the NIH Pragmatic Trials Collaboratory highlight the importance of planning for monitoring and monitoring during the trial. During the planning phase, consider how to detect unintended consequences, identify who will be responsible for intervening, and establish thresholds that would prompt action. Plans can also be created to assess imbalances in recruitment and accrual and potential lack of fidelity or adherence to the intervention. Trial leadership as well as single DSMP monitors or DSMB members must agree on what evidence requires action and/or modification of the protocol and what information and thresholds would lead to a decision to pivot or terminate the trial. During the trial, conduct repeated, rigorous checks on recruitment and accrual, sample size, and adherence and fidelity to ensure the intervention is delivered as intended. If problems arise, consider changes that allow the trial to continue to generate useful evidence. Having champions at each site and strong relationships with people who can help rectify issues is also beneficial. Because developing monitoring plans warrants special attention for PCTs, we recommend including someone who is knowledgeable about PCTs and novel research designs [4].

Case Examples

The trials used as case examples are shown in Table 1.

Funding sources

This work was supported within the National Institutes of Health (NIH) Pragmatic Trials Collaboratory through cooperative agreement U24AT009676 from the National Center for Complementary and Integrative Health (NCCIH), the National Institute on Aging (NIA), the National Institute of Mental Health (NIMH), the National Institute of Arthritis and Musculoskeletal and Skin Diseases (NIAMS), the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), the National Institute on Drug Abuse (NIDA), the NIH Office of Behavioral and Social Sciences Research (OBSSR), and the NIH Office of Disease Prevention (ODP). Demonstration Projects within the NIH Pragmatic Trials Collaboratory were supported by the following cooperative agreements with NIH Institutes: ACP PEACE (UG3AG060626, UH3AG060626), EMBED (UG3DA047003, UH3DA047003), FM-TIPS (UG3AR076387, UH3AR076387), HiLo (UG3DK118748), LIRE (UH2AT007766, UH3AR066795), NOHARM (UG3AG067593, UH3AG067593), and SPOT (UH2AT007755, UH3MH007755). The content is solely the responsibility of the authors and does not necessarily represent the official views of the NCCIH, NIA, NIMH, NIDDK, NIAMS, NIDA, OBSSR, or ODP, or the NIH.

Data Sharing

No data were generated in this manuscript, which focuses on lessons learned about monitoring clinical trials.

References

1. eCFR :: Home. https://www.ecfr.gov/#se45.1.46_1102 (accessed 3 April 2024)
2. Bunning BJ, Hedlin H, Chen JH, et al. The evolving role of data & safety monitoring boards for real-world clinical trials. *J Clin Trans Sci*. 2023;7:e179. doi: 10.1017/cts.2023.582
3. DeMets DL, Ellenberg SS. Data Monitoring Committees — Expect the Unexpected. *N Engl J Med*. 2016;375:1365–71. doi: 10.1056/NEJMra1510066 [PubMed: 27705256]
4. Ellenberg SS, Culbertson R, Gillen DL, et al. Data monitoring committees for pragmatic clinical trials. *Clinical Trials*. 2015;12:530–6. doi: 10.1177/1740774515597697 [PubMed: 26374679]

5. Zahrieh D, Croghan IT, Inselman JW, et al. Guidelines for Data and Safety Monitoring in Pragmatic Randomized Clinical Trials Using Case Studies. *Mayo Clinic Proceedings*. 2023;98:1712–26. doi: 10.1016/j.mayocp.2023.02.019 [PubMed: 37923529]
6. Simon GE, Shortreed SM, Rossom RC, et al. Principles and procedures for data and safety monitoring in pragmatic clinical trials. *Trials*. 2019;20:690. doi: 10.1186/s13063-019-3869-3 [PubMed: 31815644]
7. Ali J, Morain SR, O'Rourke PP, et al. Responding to signals of mental and behavioral health risk in pragmatic clinical trials: Ethical obligations in a healthcare ecosystem. *Contemporary Clinical Trials*. 2022;113:106651. doi: 10.1016/j.cct.2021.106651 [PubMed: 34998990]
8. Fiscella K, Sanders M, Holder T, et al. The role of data and safety monitoring boards in implementation trials: When are they justified? *J Clin Trans Sci*. 2020;4:229–32. doi: 10.1017/cts.2020.19
9. Weinfurt KP, Hernandez AF, Coronado GD, et al. Pragmatic clinical trials embedded in healthcare systems: generalizable lessons from the NIH Collaboratory. *BMC Med Res Methodol*. 2017;17:144. doi: 10.1186/s12874-017-0420-7 [PubMed: 28923013]
10. Weinfurt KP (Editor). *Rethinking Clinical Trials: A Living Textbook of Pragmatic Clinical Trials*. NIH Pragmatic Trials Collaboratory 2024.
11. Boyd AD, Gonzalez-Guarda R, Lawrence K, et al. Potential bias and lack of generalizability in electronic health record data: reflections on health equity from the National Institutes of Health Pragmatic Trials Collaboratory. *Journal of the American Medical Informatics Association*. 2023;30:1561–6. doi: 10.1093/jamia/ocad115 [PubMed: 37364017]
12. Melnick ER, Jeffery MM, Dziura JD, et al. User-centred clinical decision support to implement emergency department-initiated buprenorphine for opioid use disorder: protocol for the pragmatic group randomised EMBED trial. *BMJ Open*. 2019;9:e028488. doi: 10.1136/bmjopen-2018-028488
13. Melnick ER, Nath B, Dziura JD, et al. User centered clinical decision support to implement initiation of buprenorphine for opioid use disorder in the emergency department: EMBED pragmatic cluster randomized controlled trial. *BMJ*. 2022;e069271. doi: 10.1136/bmj-2021-069271 [PubMed: 35760423]
14. Post AA, Dailey DL, Bayman EO, et al. The Fibromyalgia Transcutaneous Electrical Nerve Stimulation in Physical Therapy Study Protocol: A Multisite Embedded Pragmatic Trial. *Physical Therapy*. 2022;102:pzac116. doi: 10.1093/ptj/pzac116 [PubMed: 36036838]
15. Jarvik JG, Meier EN, James KT, et al. The Effect of Including Benchmark Prevalence Data of Common Imaging Findings in Spine Image Reports on Health Care Utilization Among Adults Undergoing Spine Imaging: A Stepped-Wedge Randomized Clinical Trial. *JAMA Netw Open*. 2020;3:e2015713. doi: 10.1001/jamanetworkopen.2020.15713 [PubMed: 32886121]
16. Jarvik JG, Comstock BA, James KT, et al. Lumbar Imaging With Reporting Of Epidemiology (LIRE)-Protocol for a pragmatic cluster randomized trial. *Contemp Clin Trials*. Published Online First: 19 October 2015. doi: 10.1016/j.cct.2015.10.003
17. Jarvik JG, Comstock BA, James KT, et al. Lumbar Imaging With Reporting Of Epidemiology (LIRE)—Protocol for a pragmatic cluster randomized trial. *Contemporary Clinical Trials*. 2015;45:157–63. doi: 10.1016/j.cct.2015.10.003 [PubMed: 26493088]
18. Dziura J, Gilstad-Hayden K, Coffman CJ, et al. Recommendations for monitoring adherence and fidelity in pragmatic trials based on experience in the Pain Management Collaboratory. *Pain Medicine*. 2024;25:S41–8. doi: 10.1093/pm/pnae080 [PubMed: 39514878]
19. Coronado GD, Petrik AF, Vollmer WM, et al. Effectiveness of a Mailed Colorectal Cancer Screening Outreach Program in Community Health Clinics: The STOP CRC Cluster Randomized Clinical Trial. *JAMA Intern Med*. 2018;178:1174. doi: 10.1001/jamainternmed.2018.3629 [PubMed: 30083752]
20. Redmond S, Mayo Clinic NOHARM Research Team, Tilburt J, et al. Non-pharmacological Options in Postoperative Hospital-Based and Rehabilitation Pain Management (NOHARM): Protocol for a Stepped-Wedge Cluster-Randomized Pragmatic Clinical Trial. *Pain Ther*. 2022;11:1037–53. doi: 10.1007/s40122-022-00393-x [PubMed: 35657564]

21. Morain SR, Weinfurt K, Bollinger J, et al. Ethics and Collateral Findings in Pragmatic Clinical Trials. *The American Journal of Bioethics*. 2020;20:6–18. doi: 10.1080/15265161.2020.1689031
22. Finkelstein JA, Brickman AL, Capron A, et al. Oversight on the borderline: Quality improvement and pragmatic research. *Clinical Trials*. 2015;12:457–66. doi: 10.1177/1740774515597682 [PubMed: 26374685]
23. Faden RR, Kass NE, Goodman SN, et al. An ethics framework for a learning health care system: a departure from traditional research ethics and clinical ethics. *Hastings Cent Rep*. 2013;Spec No:S16–27. doi: 10.1002/hast.134 [PubMed: 23315888]
24. Morain S, Largent E. Think Pragmatically: Investigators' Obligations to Patient-Subjects When Research is Embedded in Care. *The American Journal of Bioethics*. 2023;23:10–21. doi: 10.1080/15265161.2022.2063435 [PubMed: 35435790]
25. Post AA, Dailey DL, Bayman EO, et al. The Fibromyalgia Transcutaneous Electrical Nerve Stimulation in Physical Therapy Study Protocol: A Multisite Embedded Pragmatic Trial. *Physical Therapy*. 2022;102:pzac116. doi: 10.1093/ptj/pzac116 [PubMed: 36036838]
26. Collins FS, Koroshetz WJ, Volkow ND. Helping to End Addiction Over the Long-term: The Research Plan for the NIH HEAL Initiative. *JAMA*. 2018;320:129. doi: 10.1001/jama.2018.8826 [PubMed: 29896636]
27. Lakin JR, Brannen EN, Tulskey JA, et al. Advance Care Planning: Promoting Effective and Aligned Communication in the Elderly (ACP-PEACE): the study protocol for a pragmatic stepped-wedge trial of older patients with cancer. *BMJ Open*. 2020;10:e040999. doi: 10.1136/bmjopen-2020-040999
28. Prekker ME, Driver BE, Trent SA, et al. Video versus Direct Laryngoscopy for Tracheal Intubation of Critically Ill Adults. *N Engl J Med*. 2023;389:418–29. doi: 10.1056/NEJMoa2301601 [PubMed: 37326325]
29. Platt R, Bosworth HB, Simon GE. Making Pragmatic Clinical Trials More Pragmatic. *JAMA*. Published Online First: 2 October 2024. doi: 10.1001/jama.2024.19528
30. Melnick ER, Jeffery MM, Dziura JD, et al. User-centred clinical decision support to implement emergency department-initiated buprenorphine for opioid use disorder: protocol for the pragmatic group randomised EMBED trial. *BMJ Open*. 2019;9:e028488. doi: 10.1136/bmjopen-2018-028488
31. Melnick ER, Nath B, Dziura JD, et al. User centered clinical decision support to implement initiation of buprenorphine for opioid use disorder in the emergency department: EMBED pragmatic cluster randomized controlled trial. *BMJ*. 2022;e069271. doi: 10.1136/bmj-2021-069271 [PubMed: 35760423]
32. Jarvik JG, Meier EN, James KT, et al. The Effect of Including Benchmark Prevalence Data of Common Imaging Findings in Spine Image Reports on Health Care Utilization Among Adults Undergoing Spine Imaging: A Stepped-Wedge Randomized Clinical Trial. *JAMA Netw Open*. 2020;3:e2015713. doi: 10.1001/jamanetworkopen.2020.15713 [PubMed: 32886121]
33. Redmond S, Mayo Clinic NOHARM Research Team, Tilburt J, et al. Non-pharmacological Options in Postoperative Hospital-Based and Rehabilitation Pain Management (NOHARM): Protocol for a Stepped-Wedge Cluster-Randomized Pragmatic Clinical Trial. *Pain Ther*. 2022;11:1037–53. doi: 10.1007/s40122-022-00393-x [PubMed: 35657564]
34. Simon GE, Shortreed SM, Rossom RC, et al. Effect of Offering Care Management or Online Dialectical Behavior Therapy Skills Training vs Usual Care on Self-harm Among Adult Outpatients With Suicidal Ideation: A Randomized Clinical Trial. *JAMA*. 2022;327:630. doi: 10.1001/jama.2022.0423 [PubMed: 35166800]
35. Lakin JR, Brannen EN, Tulskey JA, et al. Advance Care Planning: Promoting Effective and Aligned Communication in the Elderly (ACP-PEACE): the study protocol for a pragmatic stepped-wedge trial of older patients with cancer. *BMJ Open*. 2020;10:e040999. doi: 10.1136/bmjopen-2020-040999
36. Edmonston DL, Isakova T, Dember LM, et al. Design and Rationale of HiLo: A Pragmatic, Randomized Trial of Phosphate Management for Patients Receiving Maintenance Hemodialysis. *American Journal of Kidney Diseases*. 2021;77:920–930.e1. doi: 10.1053/j.ajkd.2020.10.008 [PubMed: 33279558]

Highlights

- Pragmatic clinical trials have distinct characteristics that should be considered when developing a monitoring plan.
- Pragmatic trials are typically embedded in clinical workflow, use existing data sources such as the electronic health record and insurance claims, and frontline clinicians are often providing the intervention (instead of dedicated research staff)
- Potential challenges include delays in data acquisition and lack of interaction between study staff and participants.
- Someone knowledgeable about pragmatic trials should help develop the monitoring plan with these considerations in mind.

Table 1.

NIH Pragmatic Trials Collaboratory Trials

Name	Setting	Intervention	Primary Outcome	Data Monitoring Plan
EMBED ClinicalTrials.gov Identifier: NCT03658642 Status: Complete	21 sites at 5 healthcare systems; 1,413,693 ED visits were assessed	user-centered clinical decision support	initiation of buprenorphine in the emergency department among patients with opioid use disorder	Data were sent to the data coordinating center every 2 weeks. The DCC monitored study data from all participating sites to ensure data quality and accuracy [30,31]. EMBED used an independent study monitor given the observational nature of our work around implementation of best practices.
FM-TIPS ClinicalTrials.gov Identifier: NCT04683042 Status: Ongoing	34 physical therapy clinics; 450 people with fibromyalgia	A TENS unit applied for 2 hours a day during physical therapy exercises and routine activity	reduction of movement-evoked pain measured on an 11-point Numeric Rating Scale from day 1 to 60 (change score)	A participant-reported change in health status excluding them from TENS, or involving a visit to the emergency room, or hospitalization prompts a Participant Experience follow-up form. If it is unclear if the event is related to TENS or study procedures is unclear, the study team follows up with the participant. The medical monitor documents relatedness and severity of the AE, information regarding the onset and stop date of the AE, and any action that was taken [25].
LJIRE[32] ClinicalTrials.gov Identifier: NCT02015455 Status: Complete	4 integrated health systems, 250,401 participants	benchmark prevalence data inserted into routine radiology reports for people who had received lumbar spine imaging	Health care utilization measured by spine-related relative value units (RVUs) within 365 days of index imaging	Two external safety officers monitored safety endpoints every six months for the duration of the study [17].
NOHARM[33] ClinicalTrials.gov Identifier: NCT04570371 Status: Ongoing	six geographically diverse locations within the Mayo Clinic Enterprise	Healing After Surgery portal guide directing patients to identify preferred nonpharmacological pain care modalities	Post-operative PROMIS pain interference and physical functioning computer adaptive tests (CAT) at 1, 2, and 3 months	Fidelity and safety of intervention delivery were monitored via the EHR [33]. Nurses' delivery of appropriate education and safe clinical use of nonpharmacological modalities were tracked in the EHR, the latter with ICD code automated surveillance.
SPOT ClinicalTrials.gov Identifier: NCT02326883 Status: Complete	4 health systems, 18 644 patients	2 low-intensity suicide outreach interventions	Time to first nonfatal or fatal self-harm	Data regarding hospital admissions, emergency department visits, and self-harm diagnoses are extracted from health system records. A National Institute of Mental Health standing DSMB reviews study progress (including blinded interim analyses regarding potential harm) three times per year [34].
ACP Peace[35] ClinicalTrials.gov Identifier: NCT03609177 Status: Ongoing	36 oncology clinics, with 30,000 patients enrolled	VitalTalk (a clinician communication skills training program) and ACP Decisions (a video library educating patients on decision-making)	Proportion of eligible patients with advance care planning documentation completed in the electronic health record.	Fidelity to the video component of the intervention will be monitored by tracking of video use (which videos are used at each clinic, playthrough rate and frequency) [35].
HiLo ClinicalTrials.gov Identifier: NCT04095039 Status: Stopped early	80-120 dialysis facilities and 4,400 participants	phosphate binder prescriptions and dietary recommendations to achieve the "Hi" serum phosphate target (≥ 6.5 mg/dL) or the "Lo" serum phosphate target (< 5.5 mg/dL).	Hierarchical composite outcome of all-cause mortality and all-cause hospitalization	HiLo delivered monthly reports to participating dietitians that provide details on achieved serum phosphate levels in the overall facility and within each individual study participant, along with local and study-wide enrollment data [36].

ED = Emergency Department; PROMIS = Patient Reported Outcomes Measurement Information System

Table 2.
Challenges and Recommendations for Monitoring in PCTs

Key Challenge	Recommendation
Data may be collected from a variety of existing sources including, for example, the electronic health record, insurance claims, death records, as well as patient-reported outcomes. There can be insufficient enrolment or biased enrollment across different demographic populations.	Conduct repeated, careful, and rigorous checks for recruitment and accrual. Create a plan for addressing imbalances, potentially by adding sites or clinics.
Clustering of patients can occur within providers and facilities.	Trials can conduct interim sample size reassessment focusing on estimates of variance or ICCs to ensure that the planned sample size is sufficient for inference.
PCT investigators often have little or no direct contact with participants	<i>A priori</i> , consider how unintended consequences could be detected
PCTs may include multiple heterogeneous sites	Early and regular adherence/fidelity monitoring is needed to determine whether intervention delivery is as intended.
PCTS are typically embedded in clinical workflow, and frontline clinicians are often providing the intervention.	Monitor adherence to the protocol (fidelity) and develop plans to increase adherence to increase adherence, such as plan-do-study-act cycles [19], re-training, or by engaging with partners, including patients, health systems leaders, and front-line clinicians.
PCTs can reveal potential incidental findings, clinical needs, or other actionable signals of distress	Determinations about responsible parties and subsequent thresholds for action for researchers and clinical staff are needed.
Data may be delayed or of poor quality due to the complex data pipeline.	Check and validate data early, hire and fund experts in IT, and engage champions at each site to establish relationships with people who can help rectify any issues.
To the amount of evidence required to drive actionable change in care varies across health systems.	If there are problems with integrity or sample size, changes should be made that allow for a pivot such that the significant investment in the trial can still lead to evidence that is useful.